

Practical #3

Practical Title:

To apply GitHub-based collaboration techniques including branching and merging

Theory:

1. What is GitHub Collaboration?

GitHub collaboration is the process where multiple developers work together on the same project using GitHub.

It allows teams to contribute, review code, manage versions, and track changes efficiently.

2. What is Branching in Git?

Branching means creating a separate version of the code to work on new features or fixes without affecting the main project.

It helps developers work independently.

3. What is Merging?

Merging is the process of combining changes from one branch into another (usually into the main branch).

4. What is Pull Request (PR)?

A Pull Request is a request to merge changes from one branch into another.

It allows team members to review, discuss, and approve code before merging.

Procedure:

Step 1: Create a GitHub Repository

Step 2: Clone Repository

Step 3: Create a New Branch

Step 4: Make Changes and Commit

Step 5: Push Branch to GitHub

Step 6: Create Pull Request on GitHub

Step 7: Merge Pull Request

Step 8: Pull Updated Code

Output Screenshots:

```
C:\Users\MRUDANG WANJARI>git clone https://github.com/viktorbezdek/awesome-github-projects
Cloning into 'awesome-github-projects'...
remote: Enumerating objects: 762, done.
remote: Counting objects: 100% (50/50), done.
remote: Compressing objects: 100% (26/26), done.
remote: Total 762 (delta 45), reused 27 (delta 24), pack-reused 712 (from 3)
Receiving objects: 100% (762/762), 1.30 MiB | 242.00 KiB/s, done.
Resolving deltas: 100% (490/490), done.
```

```
$ git remote -v
origin https://github.com/mrudangwanjari/Agent_hub.git (fetch)
origin https://github.com/mrudangwanjari/Agent_hub.git (push)
```

```
$ git merge feature1
```

```
Already up to date.
```

```
$ git checkout feature1
```

```
Switched to branch 'feature1'
```

```
$ git log
commit 04096f1e0ba8d5990147d6a4d787e13c54bb6fd7 (HEAD -> main, origin/main, origin/HEAD)
Author: mrudangwanjari <wmrudang@gmail.com>
Date: Wed Feb 11 15:14:40 2026 +0530
```

```
first commit
```

```
commit bebe430243726aaf79b11b089ee8a5b401eb81af
Merge: 4c7ecdb 0cfcf85
Author: mrudangwanjari <wmrudang@gmail.com>
Date: Tue Feb 10 11:43:05 2026 +0530
```

```
Merge branch 'main' of https://github.com/Maheen-M02/Agent_Hub
```

```
commit 4c7ecdb80744fd6df443ac74a2594bb36880d8a
Author: mrudangwanjari <wmrudang@gmail.com>
Date: Tue Feb 10 11:42:40 2026 +0530
```

```
changes through kiro
```

```
commit 0cfcf855cd7d35ee9f88616dc10a3f86adc47d79
Author: Maheen Meshram <maheenmeshram.skill102@gmail.com>
Date: Tue Feb 10 10:43:23 2026 +0530
```

```
Delete .bolt directory
```

```
commit 6c36eb1dc0d3b6294fb432d76433010a8378446a
Merge: 3c96d6b eaf166d
Author: Maheencool <maheenmeshram.2115@gmail.com>
Date: Tue Feb 10 10:42:24 2026 +0530
```

```
Merge branch 'main' of https://github.com/Maheen-M02/Agent_Hub
```

```
commit 3c96d6b88f5806761276a546f6d83e037e9893ce
Author: Maheencool <maheenmeshram.2115@gmail.com>
Date: Tue Feb 10 10:41:54 2026 +0530
```

```
changes
```

```
$ git branch
```

```
* main
```

```
$ git pull
```

```
Already up to date.
```

```
$ git checkout
```

```
Your branch is up to date with 'origin/main'.
```

Activity / Task:

Task 1:

Create a repository and collaborate with at least one teammate.



Repository Link: <https://github.com/sharvayuzade/DevOps-SEM-VI>

Task 2: Steps to Resolve Merge Conflicts

1. Pull latest changes
2. Open conflicted file
3. Identify conflict markers (<<<<<<, =====, >>>>>>)
4. Edit and fix conflicts manually
5. Add resolved file:

```
git add .
```

6. Commit changes:

```
git commit -m "Resolved merge conflict"
```

7. Push changes:

```
git push
```

Advantages of GitHub Collaboration

1. Easy teamwork and collaboration

2. Code review using Pull Requests
 3. Version tracking
 4. Parallel development using branches
 5. Integration with CI/CD tools
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Challenges

1. Merge conflicts
2. Requires proper coordination
3. Learning curve for beginners